NATURAL LANGUAGE PROCESSING

Project Title

FINROOT: An Explainable NLP-Based Financial Complaint Root-Cause and Resolution Engine

Programme	BBA (Artificial Intelligence) – Section B
Academic Year	2026-27
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1. Abstract

Banks and FinTech companies receive large volumes of free-text customer complaints that must be read, interpreted and routed by support staff. Conventional classifiers return only a broad category and give no evidence or next step. This project presents **FINROOT**, a decision-support prototype that converts a complaint narrative into a root-cause category, a confidence score, word-level evidence and a retrieved resolution path, with a human-review fallback when the system is unsure.

The project uses complaint narratives from the U.S. CFPB Consumer Complaint Database (July 2026 export). A 10,002-narrative data package was prepared, with duplicate tracking and a 1,200-narrative entity/event annotation set. The modelling notebook uses a labelled subset of 288 narratives (230 train / 58 test). Two classifiers over provisional root-cause labels were compared: TF-IDF (1–2-grams) with Logistic Regression, and sentence embeddings (all-MiniLM-L6-v2) with Logistic Regression. LIME is used for explanation and a curated knowledge base for resolution retrieval, with a human-review fallback for low-confidence cases.

On the test set, the embedding model reached 0.71 accuracy (macro-F1 0.35) and TF-IDF 0.66 (macro-F1 0.24), compared with 53.4% for always predicting the majority class. Identity-theft complaints are recognised very well (F1 up to 1.00). The working prototype demonstrates the complete path from complaint text to root cause, evidence and routed outcome, and identifies clear next steps for scaling it to the full corpus.

2. Introduction

2.1 Background and context

Customers complain about failed transactions, missing refunds, duplicate charges, incorrect fees, unauthorised activity and collection practices. Complaints are written informally, often contain spelling errors, and frequently mix several issues in one message. Manual triage is slow, and different agents may read the same complaint differently.

2.2 Motivation

A label such as “payment issue” tells an agent little about what actually went wrong or what to do next. We wanted a system that goes further: identify the likely underlying problem, show which words in the complaint support that diagnosis, and point to a reviewed procedure – while being explicit when it cannot be trusted. The CFPB database offers a large public corpus of real consumer narratives, which makes it a realistic testbed even though it is U.S.-centred.

2.3 Role of NLP

The input is unstructured text, so every stage is an NLP task: text normalisation, information extraction (amounts, dates, products, events), text representation (TF-IDF and sentence embeddings), supervised text classification, post-hoc explainability (LIME) and category-to-procedure retrieval. FINROOT is deliberately an NLP pipeline rather than a chatbot: the model outputs are inspectable and measurable at each step.

3. Problem Statement and Objectives

Problem statement. How can NLP convert unstructured financial complaints into explainable problem diagnoses and relevant resolution procedures, while flagging uncertain cases for human review?

Objectives

1. Build a clean, auditable corpus of real financial complaint narratives with duplicate and leakage controls.
2. Annotate a representative subset with entities and events and implement a baseline information-extraction module.
3. Train and compare a TF-IDF baseline with a semantic-embedding classifier for root-cause prediction.
4. Explain individual predictions with LIME.
5. Link predicted categories to a curated resolution knowledge base with a safe fallback.
6. Evaluate with accuracy, precision, recall, F1 and a confusion matrix; audit robustness across writing styles; and report failures honestly.

4. Dataset Description

Source. CFPB Consumer Complaint Database (Consumer Financial Protection Bureau, U.S.), July 2026 export, filtered to records that contain a consumer complaint narrative. Link: https://www.consumerfinance.gov/data-research/consumer-complaints/

4.1 Prepared data package

Table 1: Summary of the FINROOT primary corpus (data package)

Property	Value
Records (narratives)	10,002 (20 columns)
Unique normalised narratives	9,379; 792 rows are duplicate narratives
Financial products / CFPB issues / sub-issues	11 / 82 / about 190
Date received	1–29 July 2026; all submitted via Web
Narrative length	Mean 245.7 words (median 187, max 4,897); mean 13.9 sentences
Language	English (U.S. consumers)
Key fields	Complaint ID, Product, Sub-product, Issue, Sub-issue, Consumer complaint narrative, Company, State, narrative_clean, narrative_hash, duplicate flags, length statistics, root_cause

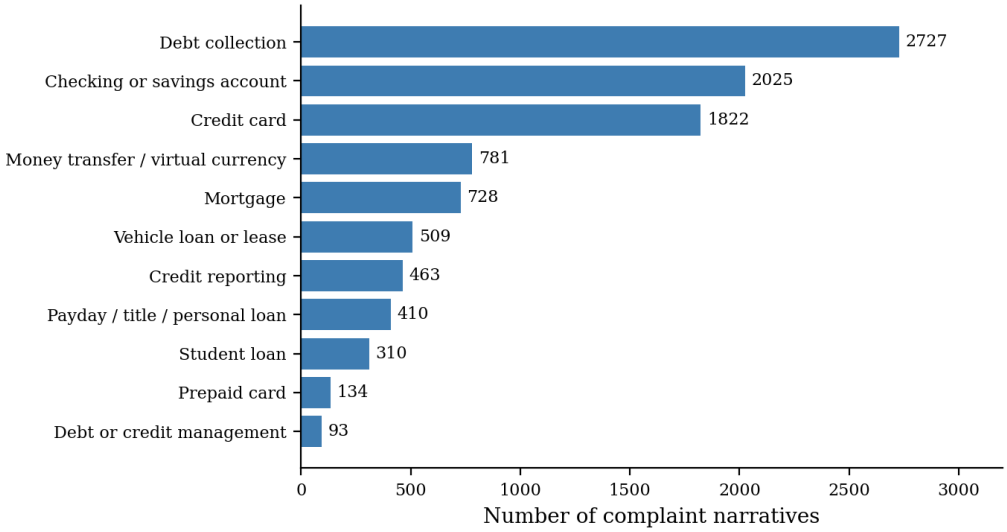


Figure 1: Distribution of complaint narratives across the 11 CFPB product categories in the data package (n = 10,002).

Annotation set. A representative review set of 1,200 narratives was annotated with 7,819 spans (4,799 entity and 3,020 event spans; Table 2). Annotations were produced by a rule-based extractor followed by a rule-based adjudication pass with exact-span validation. They are recorded as AI_REVIEWED, **not human-verified gold**, and no gold-based metric is reported here.

Table 2: Most frequent annotation labels in the 1,200-narrative review set

Label	Spans	Label	Spans
FINANCIAL_PRODUCT	1,836	LOCATION	253
PERSON_OR_PARTY	1,687	MONEY_CREDITED	233
MONEY_SENT	858	ACCOUNT	114
AMOUNT	839	UNAUTHORIZED_ACTIVITY	89
RECEIPT_OUTCOME	723	ACCESS_EVENT	86
CHARGED	703	TRANSACTION_ID	63
MONEY_DEBITED	281	Other (refund, payment, date)	54

4.2 Data actually used for modelling

The modelling notebook reads the CSV with permissive settings (quote handling disabled and malformed lines skipped) to handle unclosed quotes in the file. After selecting the text column (narrative_clean) and the label column (root_cause), 377 parsed rows had no label and were removed, and classes with fewer than two examples were dropped so that stratified splitting was possible. This left **288 labelled narratives**, split 80/20 (random_state = 42, stratified) into **230 training and 58 test** narratives. The full 10,002-narrative package is available for scaling the experiments.

Labels. The root_cause column has eight classes in the modelling subset (seven appear in the test set). CFPB Issue and Sub-issue are source metadata, not verified root causes, so these root_cause labels are treated as **provisional (silver) labels** that have not been checked against narratives by human annotators. A separate twelve-category taxonomy (RC01–RC12, Annexure A) was drafted but is not yet applied.

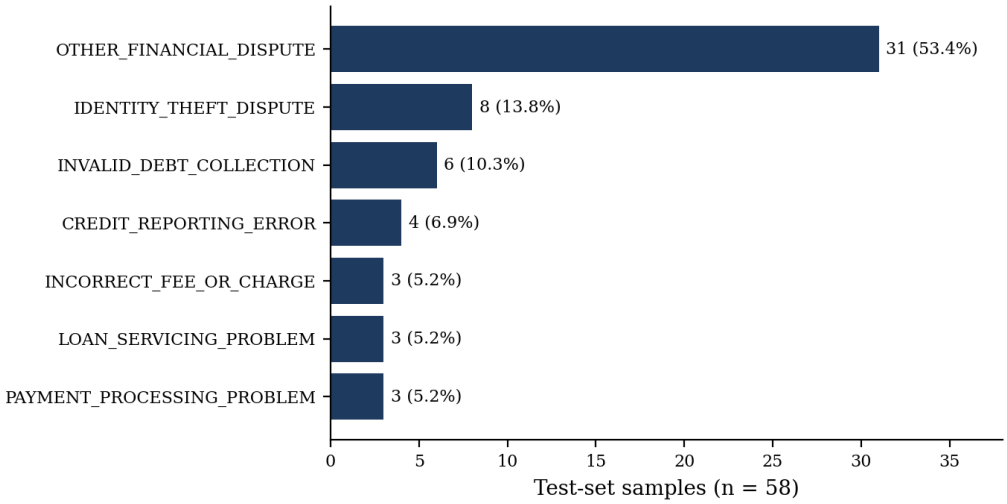


Figure 2: Class distribution of the root-cause labels in the test split (n = 58).

The class distribution is highly imbalanced: OTHER_FINANCIAL_DISPUTE accounts for 31 of 58 test narratives (53.4%) and four classes have only three examples each.

5. Data Preprocessing

Preprocessing was intentionally light, because punctuation, wording and masked-number patterns can themselves be evidence for later NLP steps.

- **Unicode, line-ending and whitespace normalisation**, followed by outer trimming, producing the narrative_clean field. The original narrative is preserved for auditability.
- **No aggressive normalisation:** stop-words, case and punctuation were not removed from the stored text; masked tokens (e.g., XXXX placeholders), legal references and spelling noise were kept. TF-IDF lowercases internally; the sentence-embedding model receives the raw cleaned text.
- **Filtering:** only records with a non-empty narrative were retained. In the modelling file, no text values were missing, but 377 rows had no label and were dropped before training.
- **Duplicate control:** in the data package, a hash of the normalised narrative (narrative_hash) flags 792 duplicate rows. In the modelling notebook, an exact-text check found **14 duplicate narratives among the 288 labelled rows**, none with conflicting labels; removing them leaves 274 unique narratives. This deduplication was performed after the train/test split and was not used to retrain the models.
- **Leakage control:** company response, timely-response and other post-complaint fields were retained in the corpus but excluded from model inputs, so the classifier uses only text available at intake.

- **Not handled:** code-mixed or non-English text, URLs/hashtags/emojis (largely absent in this formal web-form corpus), and spelling correction.

6. Feature Representation / Embeddings

Two representations were compared so that the benefit of semantic information could be measured rather than assumed.

- **TF-IDF (baseline).** Word unigrams and bigrams, vocabulary capped at 10,000 features. TF-IDF is fast, interpretable and a strong baseline for topic-like classification, but it cannot connect differently worded descriptions of the same problem (e.g., “deducted” and “cut from my account”).
- **Sentence embeddings.** The pretrained all-MiniLM-L6-v2 Sentence-Transformer encodes each complaint into a dense vector so that semantically similar complaints lie close together, which is the property needed for robustness to paraphrase and informal wording. Each complaint is encoded as a single vector without fine-tuning.

One limitation of this choice is that MiniLM truncates long inputs (the model’s maximum sequence length is far shorter than the longest narratives, which reach 4,897 words), so only the opening part of long complaints is represented.

7. Proposed Methodology / System Architecture

FINROOT follows the pipeline Complaint → Understanding → Root Cause → Evidence → Resolution (Figure 3). A confidence gate and a “no verified procedure” fallback ensure that uncertain or unsupported cases are routed to a human instead of receiving an invented answer.

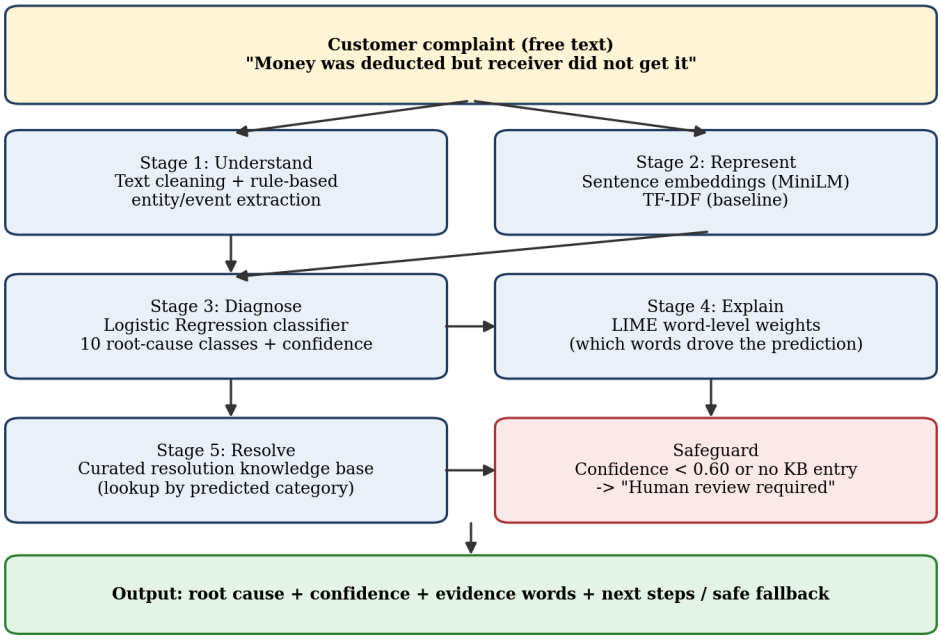


Figure 3: End-to-end FINROOT workflow from complaint text to root cause, evidence and resolution (or human review).

The workflow in this project was carried out in three parts: (i) data and information extraction (corpus, annotation set, extractor, taxonomy draft); (ii) modelling (TF-IDF and embedding

classifiers, evaluation); and (iii) explanation, resolution retrieval and the robustness audit, which are combined in the function finroot_analyze().

8. NLP Model / Technique Used

8.1 Root-cause classifiers

Both classifiers use multinomial Logistic Regression (scikit-learn, max_iter = 1000). The data was split 80/20 with train_test_split (random_state = 42, stratified by label), giving 230 training and 58 test narratives. No class weighting, resampling, cross-validation or hyper-parameter tuning was applied.

Table 3: Experimental settings

Setting	Baseline	Semantic model
Representation	TF-IDF, 1–2-grams, max 10,000 features	all-MiniLM-L6-v2 sentence embeddings
Classifier	Logistic Regression (max_iter = 1000)	Logistic Regression (max_iter = 1000)
Split	80/20 stratified, seed 42	Same split
Train / test size	230 / 58	230 / 58

8.2 Information extraction

A rule-based, auditable extractor identifies currency amounts, dates, financial-product mentions and transaction events (e.g., MONEY_SENT, MONEY_DEBITED, REFUND_NOT_RECEIVED) through regular expressions with unit tests. It is a baseline, not a production NER model, and its output was used to build the annotation set rather than as classifier input.

8.3 Explainability

LIME (LimeTextExplainer) perturbs the complaint by removing words, observes how the embedding-model probabilities change, and fits a local linear model to estimate each word's contribution (8 features per explanation).

8.4 Resolution retrieval and safeguards

A dictionary-based knowledge base maps a category to a problem description, ordered steps and an escalation note. get_resolution() returns "Human review required" for any category without a reviewed entry. In addition, if the prediction confidence is below 0.60, the output is replaced by a human-review message. Resolution text is hand-written and is never generated by a model.

9. Implementation

- **Language and libraries:** Python in Google Colab; pandas and NumPy (data), scikit-learn (TF-IDF, Logistic Regression, metrics), sentence-transformers (all-MiniLM-L6-v2), LIME (explanations), matplotlib and seaborn (plots).
- **Modules:** data/QC package; annotation guidelines and review set; baseline extractor with unit tests; TF-IDF and embedding classifiers; predict_root_cause() (label + maximum class probability); explain_complaint(); get_resolution(); finroot_analyze(),

which chains prediction, resolution retrieval and explanation; a knowledge-base coverage check; a duplicate check; and an audit routine that scores writing-style test cases.

- **Interface:** the prototype runs as a notebook workflow; no standalone web application has been built.

Worked example. For the complaint “Money was deducted but receiver did not get it”, finroot_analyze() returned the root cause OTHER_FINANCIAL_DISPUTE with confidence 0.571. No reviewed resolution exists for that category and the confidence is below 0.60, so the system sends the case to human review rather than offering a procedure. The LIME explanation is shown in Figure 4.

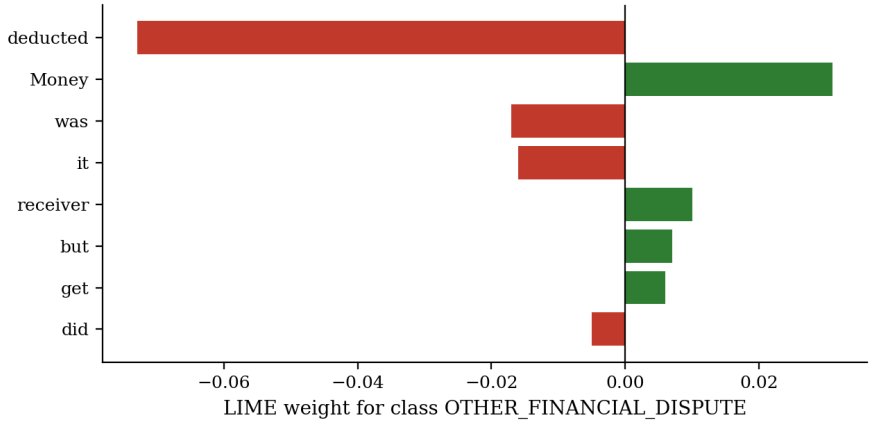


Figure 4: LIME word weights for the predicted class OTHER_FINANCIAL_DISPUTE (values taken from the notebook output).

The word “deducted” has the largest weight (−0.073, i.e., it pushes the model away from the catch-all class), which is sensible, but all weights are small, so the evidence behind this prediction is weak.

10. Results and Evaluation

All results below come from a single run on the **58-narrative test set** (one prediction equals about 1.7 percentage points), so they should be read as indicative only.

10.1 Overall performance

Table 4: Overall test-set performance (n = 58)

Metric	Majority-class rate	TF-IDF + LR	MiniLM + LR
Accuracy	0.534	0.66	0.71
Macro precision	–	0.23	0.38
Macro recall	–	0.27	0.36
Macro F1	–	0.24	0.35
Weighted F1	–	0.53	0.60

The embedding model scores higher than the TF-IDF baseline on every aggregate metric (accuracy 0.71 vs 0.66; macro-F1 0.35 vs 0.24), which is in the direction we expected for

semantic representations. The gap, however, corresponds to three test narratives (41 vs 38 correct), which is too small to count as evidence of a real difference. Always predicting OTHER_FINANCIAL_DISPUTE would already give 53.4% accuracy, so the models add 13 (TF-IDF) and 18 (MiniLM) percentage points; macro-F1 is the more honest summary and is low for both.

10.2 Per-class results

Table 5: Per-class precision (P), recall (R) and F1 on the test set (TF = TF-IDF + LR; ME = MiniLM embeddings + LR)

Class	n	P-TF	R-TF	F1-TF	P-ME	R-ME	F1-ME
CREDIT_REPORTING_ERROR	4	0.00	0.00	0.00	1.00	0.50	0.67
IDENTITY_THEFT_DISPUTE	8	1.00	0.88	0.93	1.00	1.00	1.00
INCORRECT_FEE_OR_CHARGE	3	0.00	0.00	0.00	0.00	0.00	0.00
INVALID_DEBT_COLLECTION	6	0.00	0.00	0.00	0.00	0.00	0.00
LOAN_SERVICING_PROBLEM	3	0.00	0.00	0.00	0.00	0.00	0.00
OTHER_FINANCIAL_DISPUTE	31	0.61	1.00	0.76	0.65	1.00	0.78
PAYMENT_PROCESSING_PROBLEM	3	0.00	0.00	0.00	0.00	0.00	0.00

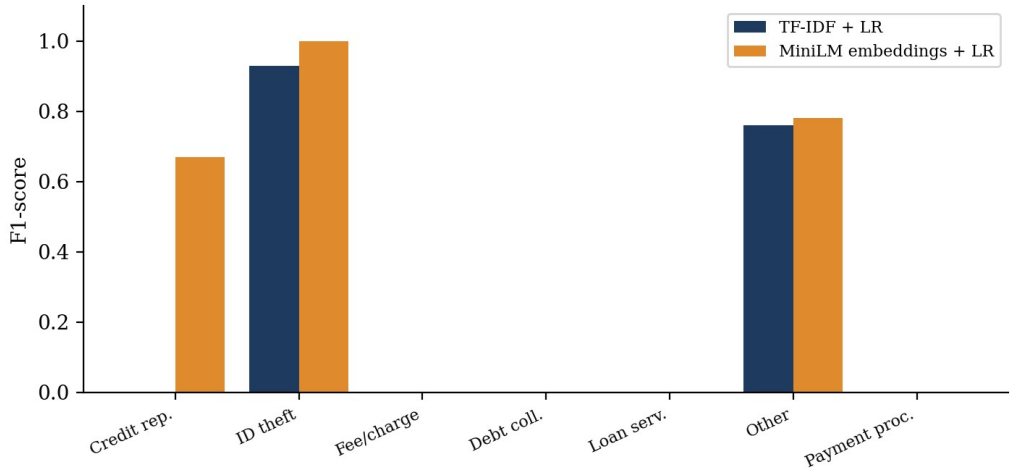


Figure 5: Per-class F1-score of the two classifiers on the 58-narrative test set.

Only two behaviours are learned: identity-theft complaints are recognised well (F1 0.93 and 1.00; these narratives contain distinctive vocabulary), and the catch-all class is always found (recall 1.00) at the cost of low precision. Four of the seven test classes have F1 = 0 for both models; credit-reporting errors are recovered only by the embedding model (recall 0.50). LOAN_PAYOFF_PROBLEM was seen in training but has no test example.

10.3 Confusion matrix

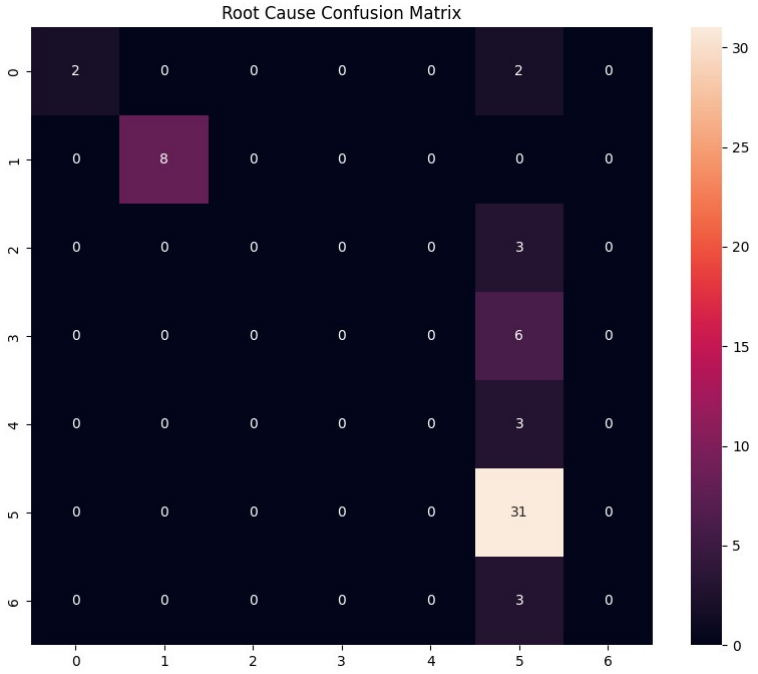


Figure 6: Confusion matrix of the MiniLM + Logistic Regression model (rows: true class, columns: predicted class; indices 0–6 follow the order of Table 5).

Every error in the matrix is a prediction of OTHER_FINANCIAL_DISPUTE (index 5). Of the 27 test complaints that are not OTHER, the model classified 10 correctly (2 credit-reporting and 8 identity-theft) and sent the remaining 17 to the catch-all class, including all 6 debt-collection and all 3 fee, loan-servicing and payment-processing cases.

10.4 Robustness audit across writing styles

Four short complaints describing the same “money deducted but recipient not credited” situation were written in different styles and passed through the MiniLM model (Table 6).

Table 6: Writing-style audit (expected label FAILED_TRANSFER)

Style	Text	Predicted	Confidence
Formal	The amount was debited but the beneficiary did not receive it.	OTHER_FINANCIAL_DISPUTE	0.564
Informal	Money got cut from my account but my friend didn’t get it.	OTHER_FINANCIAL_DISPUTE	0.700
Noisy	money debtd frm acc bt recvr didnt gt it	OTHER_FINANCIAL_DISPUTE	0.554
Hinglish	Mere account se paise deduct ho gaye but receiver ko nahi mile.	OTHER_FINANCIAL_DISPUTE	0.560

All four cases were marked incorrect (0/4). This number must be read with care: FAILED_TRANSFER is **not** one of the labels the classifier was trained on, so a correct match was impossible by construction. The audit is nevertheless informative: the model chose the catch-all class for every style, so it does not separate the transfer-failure situation from other disputes at all. Three of the four confidences (0.554–0.564) fall below the 0.60 gate and would be sent to human review; the informal case (0.700) would pass the gate with a wrong label.

10.5 Resolution retrieval

The knowledge base contains two entries (FAILED_TRANSFER and REFUND_NOT_RECEIVED). Calling get_resolution("FAILED_TRANSFER") returns the four-step procedure and escalation note, and an unknown category returns a "Human review required" message, as intended. A coverage check in the notebook shows, however, that **none of the eight classifier labels has a procedure** and both knowledge-base entries are unused, because their keys do not exist in the classifier's label space. Resolution retrieval is therefore demonstrated as a mechanism only; no resolution-quality metric is reported.

11. Error Analysis / NLP Failure Analysis

1. Dominance of the catch-all class. All 17 errors of the embedding model are predictions of OTHER_FINANCIAL_DISPUTE (Figure 6). The class makes up about half of the data, the labels are provisional, and no class weighting or resampling was applied.

2. Small classes. INCORRECT_FEE_OR_CHARGE, INVALID_DEBT_COLLECTION, LOAN_SERVICING_PROBLEM and PAYMENT_PROCESSING_PROBLEM have too few examples (3–6 in the test set) for a linear model to separate them; a larger training set is the natural remedy.

3. Paraphrase, noise and language. In the audit, formal, informal, noisy and Hinglish versions of the same complaint all received the same label. The model is trained on formal U.S. English, so Hinglish and abbreviations such as "frm" are out of scope. The 0.60 confidence gate flags three of the four cases for human review.

4. Explanation strength. LIME weights for the worked example are small (largest magnitude 0.073); the explanation shows what the model reacted to, not proof that its reasoning is correct.

5. Duplicates. 14 of the 288 labelled narratives are exact duplicates and the split was made before deduplication, so a few identical complaints may appear in both train and test sets.

12. Ethical Considerations and Limitations

- **Label validity.** The root-cause labels are provisional and were not human-verified; the 7,819 annotations are AI/rule-adjudicated, not human gold. Results measure agreement with these labels, not with true root causes.
- **Sample size.** The reported metrics rest on 58 test narratives and are indicative only.
- **Geographic and cultural scope.** The data are U.S. consumer complaints. The system must not be presented as validated for Indian banking or UPI complaints, and it does not handle Hinglish or other languages.
- **Privacy.** CFPB narratives are public and redacted by the source, but may still contain personal details or masked identifiers. The corpus was used only for research; no attempt was made to identify individuals.
- **Fairness and imbalance.** Customers whose problems fall in rare or minority categories are systematically less well served, because those categories are almost never predicted. Per-group fairness analysis was not performed.

- **Harmful or incorrect advice.** Resolution steps must not be invented by a model. FINROOT retrieves only from a curated knowledge base, which needs review against each provider's official procedure and currently covers none of the classifier's labels.
- **Over-confidence and over-reliance.** Predicted probabilities are not calibrated, and a wrong prediction can still pass the 0.60 gate. FINROOT is decision support: it should flag patterns for investigation, not prove fault by a bank, and uncertain cases should go to a human.
- **Explanation limits.** LIME explanations are local approximations and can vary between runs.

13. Conclusion

We developed an end-to-end explainable complaint-analysis pipeline: a 10,002-narrative CFPB data package with quality controls, a 1,200-narrative annotation set and rule-based extractor, two root-cause classifiers, LIME explanations, a resolution knowledge base with a safe human-review fallback, a coverage check and a writing-style audit. The working prototype demonstrates the full chain from complaint text to a routed outcome.

The objectives were met as a working pipeline. On the 58-narrative test set, MiniLM embeddings reached 0.71 accuracy and 0.35 macro-F1 versus 0.66 and 0.24 for TF-IDF, with strong performance on identity-theft complaints. The main learning is that data volume and label quality drive results more than the choice of representation: the next steps are scaling to the full corpus, human-verified labels and reviewed resolution procedures for every category.

14. Future Scope

- **Scale up:** train on the full labelled corpus and report cross-validated results with confidence intervals.
- **Human gold labels:** review the 1,200-narrative set, measure inter-annotator agreement and report entity and root-cause metrics against real gold data.
- **Apply the RC01–RC12 taxonomy** and split the OTHER class into meaningful categories; consider multi-label prediction for multi-issue complaints.
- **Imbalance handling and leakage control:** class weights, resampling and tuned thresholds; deduplicate (or group by narrative hash) before splitting.
- **Stronger models:** fine-tuned transformers (e.g., DistilBERT/RoBERTa) with probability calibration.
- **Close the retrieval gap:** align knowledge-base keys with model labels, add reviewed procedures for each label, and evaluate retrieval quality with manual review.
- **Indian context:** collect UPI and RBI-ombudsman-style complaints, add Hinglish/code-mixed data and spelling robustness, and re-run the audit with a larger and correctly specified test set.
- **Interface and monitoring:** a simple web UI for agents and clustering of similar complaints to surface recurring operational issues.

15. References

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16. GitHub Repository

- GitHub Repository Link: **[Insert final repository URL here before submission]**

The repository should contain the notebook (nlpproject.ipynb), the FINROOT data package (corpus, annotation files, taxonomy, data dictionary, QC report), the baseline extractor and its unit tests, and a README with run instructions.

17. Team Contribution

Table 7: Contribution of each team member

Team Member	PRN	Specific Contribution
Manbhavna Khanna	25030422067	Data and information extraction: CFPB data acquisition, filtering and light normalisation, duplicate/leakage controls, data dictionary and QC, 1,200-narrative annotation set and guidelines, baseline extractor with unit tests, taxonomy draft.
Tulip Salaria	25030422127	Modelling and evaluation: train/test split, TF-IDF and MiniLM embedding classifiers, classification reports, confusion matrix, per-class analysis and error analysis.
Swasti Sinha	25030422122	Explanation and resolution: LIME explanations, resolution knowledge base and retrieval with confidence gate and fallback, finroot_analyze() pipeline, writing-style audit.

All members contributed to planning, discussion of results and writing of this report.

18. Annexure

Annexure A: FINROOT root-cause taxonomy V1 (draft)

The taxonomy below was proposed for review and is not the label set used by the trained classifier (which uses eight provisional labels). The decision principle is to choose the most specific category supported by the complaint, and RC12 when evidence is insufficient.

Table A1: Draft root-cause taxonomy

ID	Name	Definition
RC01	Unauthorized activity	Unrecognised or unauthorised transaction or activity.
RC02	Duplicate transaction or charge	Same transaction, payment or charge appears more than once.
RC03	Incorrect amount or fee	Amount, fee, interest or charge is incorrect or unexpected.
RC04	Missing or failed transaction	Expected transfer, payment, deposit or withdrawal did not complete.
RC05	Refund failure or delay	Refund is missing, delayed, incorrect or not credited.
RC06	Access or authentication problem	Cannot access an account or service; authentication or verification issues.
RC07	Account servicing or process failure	Account maintenance, servicing, communication or handling problems not covered elsewhere.
RC08	Payment processing problem	Processing, posting, routing or status problems not better captured elsewhere.
RC09	Credit reporting or data accuracy	Inaccurate or incomplete credit-report data or reporting activity.
RC10	Debt or collection handling	Collection conduct, debt validity, debt handling or collection communication.
RC11	Fraud or scam concern	Suspected fraud or scam without enough evidence for RC01.
RC12	Other / insufficient evidence	Not enough evidence for any other category.

Annexure B: Resolution knowledge-base entries (prototype)

Table B1: Curated resolution procedures (to be verified against official provider guidance)

Category key	Procedure
FAILED_TRANSFER	1) Check the transaction status and reference number. 2) Verify whether the recipient was credited. 3) Check the applicable reversal or settlement timeline. 4) If unresolved, follow the provider's dispute process; escalate through the payment provider's official support channel.
REFUND_NOT_RECEIVED	1) Check the refund status with the merchant or provider. 2) Verify the stated processing timeline. 3) Compare the expected date with the actual status. 4) If overdue, raise a complaint through the official channel.

Annexure C: Data and evaluation caveats

- The reported classification scores come from a single random stratified split (seed 42) of 288 labelled narratives (230 train / 58 test), without cross-validation or confidence intervals.

- No gold-standard metrics are reported for entity/event extraction because the annotations are rule/AI-adjudicated and human verification is pending.
- Eight labels exist in the training data (CREDIT_REPORTING_ERROR, IDENTITY_THEFT_DISPUTE, INCORRECT_FEE_OR_CHARGE, INVALID_DEBT_COLLECTION, LOAN_PAYOFF_PROBLEM, LOAN_SERVICING_PROBLEM, OTHER_FINANCIAL_DISPUTE, PAYMENT_PROCESSING_PROBLEM); LOAN_PAYOFF_PROBLEM has no test example, so the confusion matrix indices 0–6 correspond to the other seven labels in this order.
- Model-label and knowledge-base coverage: 0 of 8 classifier labels have a reviewed resolution; the 2 knowledge-base keys (FAILED_TRANSFER, REFUND_NOT_RECEIVED) are unused.